

Full name: Ha Hoang PhucMSSV: 222683Class: DH22TIN07LESSON 1:  
Library management system-Identify candidate classes (Books, Readers,  
Vouchers, Employees, Publishers)+ Book: main entity, managing book  
information.

- + DocGia: person who borrows books.
- + PhieuMuon: records borrowing/returning.
- + NhanVien: system manager (librarian, warehouse keeper...).
- + NhaXuatBan: book source information.

- Properties and behavior (methods) of each class+

SachAttributes: maSach (PK, string), nameSach

(string), tacGia (string),namXuatBan (int), manNXB

(FK), soLuongTon (int), trangThai (string: "Yes

```
oBehavior: addSach(), capNhatSach(), deleteSach(),  
saveSach(), capNhatTonKho () .
```

+ DocGiaoAttributes: maDocGia (PK, string), hoTen (string), birth date (date),  
diaChi(string), soDienThoai (string), ngayDangKy (date), gioiTinh (boolean).

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oBehavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon().+
```

PhieuMuonoAttributes: maPhieu (PK, string), maDocGia (FK), maSach

(FK),FK, you're (date), you're (date), you're(date), TinhTrang

(string: "On loan"/"Paid"/"Overdue").

```
oBehavior: createPhieuMuon(), searchSach(),
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```
saveTraQuaHan(), crystalPhieuPhat().+ NhanVienoAttributes:
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maNhanVien (PK, string), hoTen (string), chucVu

(string),ngaySinh (date), soDienThoai (string).

```
oBehavior: dangNhap(), quanLyDocGia(), quanLyMuonTra(),
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```
quanLySach() .
```

+ NhaXuatBanoAttributes: maNXB (PK, string), nameNXB (string), diaChi (string),  
email(string).

```
oBehavior: addNXB(), capNhatNXB() .
```

- Relationships□Books and Publishers: 1 book –

1 Publishing House

- +Readers and Vouchers: 1 reader – many borrowing slips
- +Books and Borrowing Tickets: 1 book – many borrowing coupons
- +Employees and Loan Vouchers: 1 employee – many vouchers

## **LESSON 2: Online sales management system**

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- ΔανγκψΤαιΚηαν (Χυστομερ ρεγιστρατιον).
- Δανγκηαπ (Σιγν ιν).
- ΣεεΣανΠηαμ (Σεαρχη / ζιεω προδυχτ λιστ).
- ΤηεμζαοΓιοΗανγ (Αδδ το χαρτ).
- ΔατΗανγ (Ορδερ).
- ΤηανηΤοαν (Ονλινε παψμεντ / ΧΟΔ).
- ζιεω ΔονΗανγ (ζιεω ορδερ ηιστορψ).

Admin / NguoiQuanLy:

- ΘυανΛψΣανΠηαμ (Αδδ/Εδιτ/Δελετε προδυχτσ).
- ΘυανΛψΔονΗανγ (ζιεω/Χονφιρμ/Χανχελ ορδερ).
- ΘυανΛψΚηαχηΗανγ (Χυστομερ ινφορματιον μαναγεμεντ).
- ΘυανΛψΤηανηΤοαν (Παψμεντ προχεσσινγ, ρεφυνδ ιφ νεχεσσαρψ).

Extend / Include:

- ΤηανηΤοαν ινχλυδεσ ΞαχΝηανΔονΗανγ.
- ΔατΗανγ εξτενδσ ΤηανηΤοαν (ιφ παψινγ ονλινε).

- From use case, find the necessary classes

- +Customers (Customer): from register, view orders.
- +SanPham (Product): product management.
- +GioHang (Cart): temporarily store products before ordering.
- +DonHang (Order): from order, payment.
- +ChiTietDonHang (OrderDetail): product details in the order.
- +ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- +Admin or NguoiQuanLy (inherits from User if there is a user system).

- Analyze relationships between classes (specify inheritance if any) □ KhachHang and DonHang: 1 customer – many orders □ DonHang and ChiTietDonHang: 1 order – many order details (Composition) □ ChiTietDonHang and SanPham: many single details – 1 product □ DonHang and ThanhToan: 1 order – 1 payment □ User (abstract class) ←inherit ←KhachHang and Admin (KhachHang and Admins are all the same type of User, sharing the same account, password, and full name).

### Lesson 3: Convert class diagrams into data tables

#### 1.MonHoc Table

| No | School name | Data Types   | Bind |    |          |        |
|----|-------------|--------------|------|----|----------|--------|
|    |             |              | PK   | FK | Not NULL | Unique |
| 1  | maMon       | Nvarchar(10) | x    |    | x        | x      |
| 2  | tenMon      | Nvarchar(50) |      |    | x        |        |
| 3  | soTinChiLT  | Int          |      |    | x        |        |
| 4  | soTinChiTH  | Int          |      |    | x        |        |

#### 2.Science Table

| No | School name  | Data Types   | Bind |    |          |        |
|----|--------------|--------------|------|----|----------|--------|
|    |              |              | PK   | FK | Not NULL | Unique |
| 1  | maKhoaHoc    | Nvarchar(10) | x    |    |          | x      |
| 2  | name KhoaHoc | Nvarchar(50) |      |    | x        |        |
| 3  | rightTao     | Date         |      |    | x        |        |